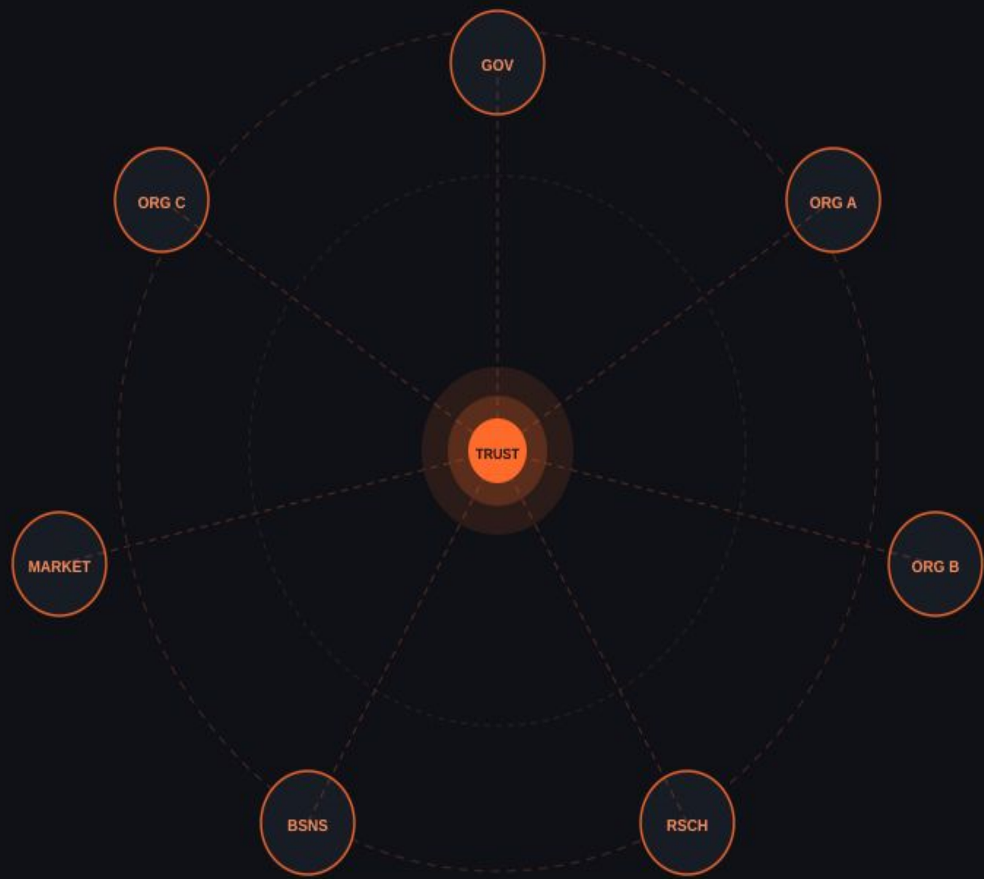
DataSpaces

Implementing the Marketplace

Fontys ICT · Lectoraat Interactive Design

Research Project · April 2026

How can we implement and configure a Marketplace inside a DataSpace?



What is a DataSpace?

A decentralized ecosystem where organisations share data under agreed rules — while retaining full ownership.

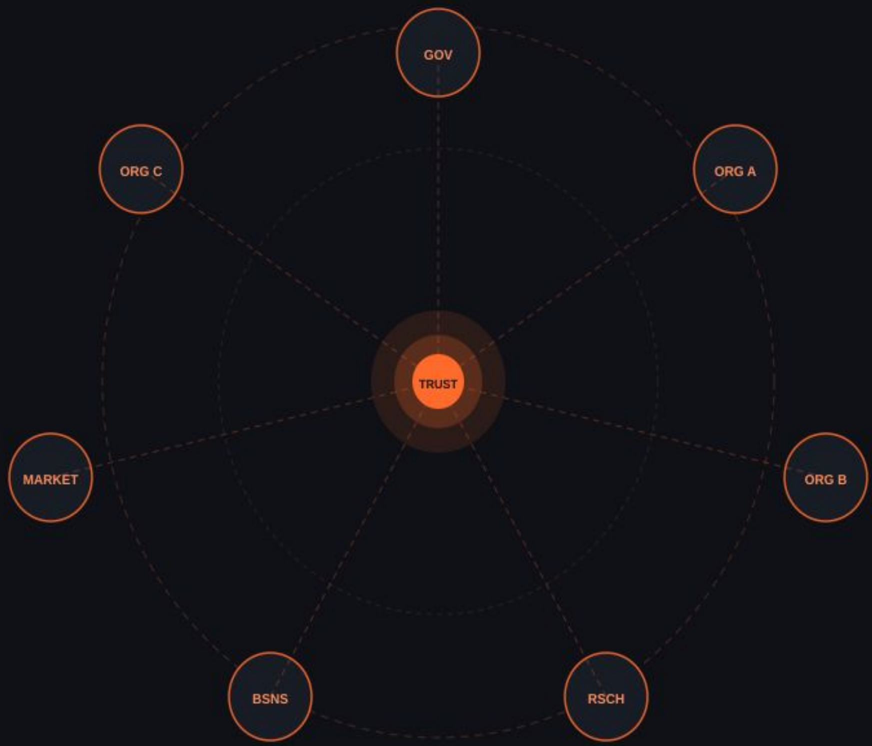
Data Sovereignty	Trust Framework	Interoperability
Own your data	Agreed rules	Open standards

Data stays at source

Providers never upload. Access is granted via secure APIs.

Consent-based access

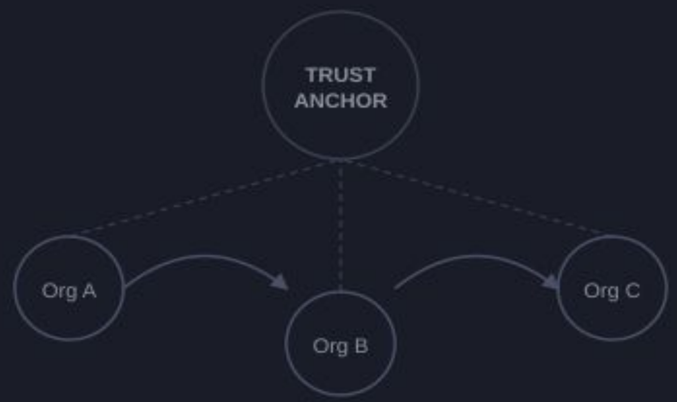
Every transaction requires explicit agreement.



DataSpace vs Marketplace

Two different concepts — this project bridges them.

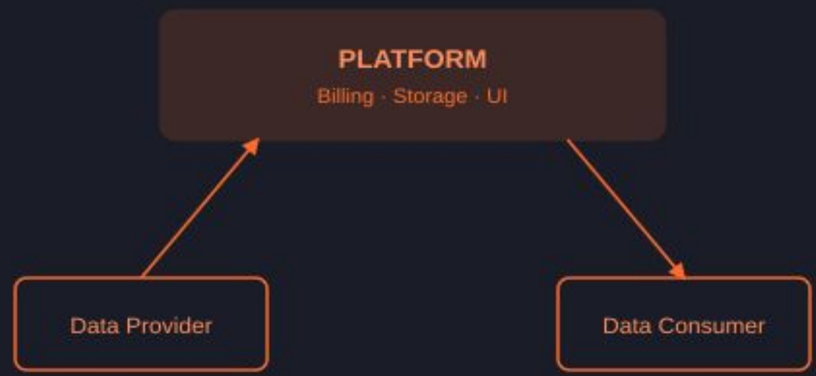
DataSpace — Decentralized



Data stays at source

You own your data

Marketplace — Centralized



Platform-mediated flow

Platform owns the flow

Project Background

Research Question

How can we implement and configure the Marketplace in a DataSpace?

Challenge

Within Data Spaces, the Marketplace acts as a facilitating layer. Implement and configure it across real use cases, addressing all DataSpace elements.

Scope

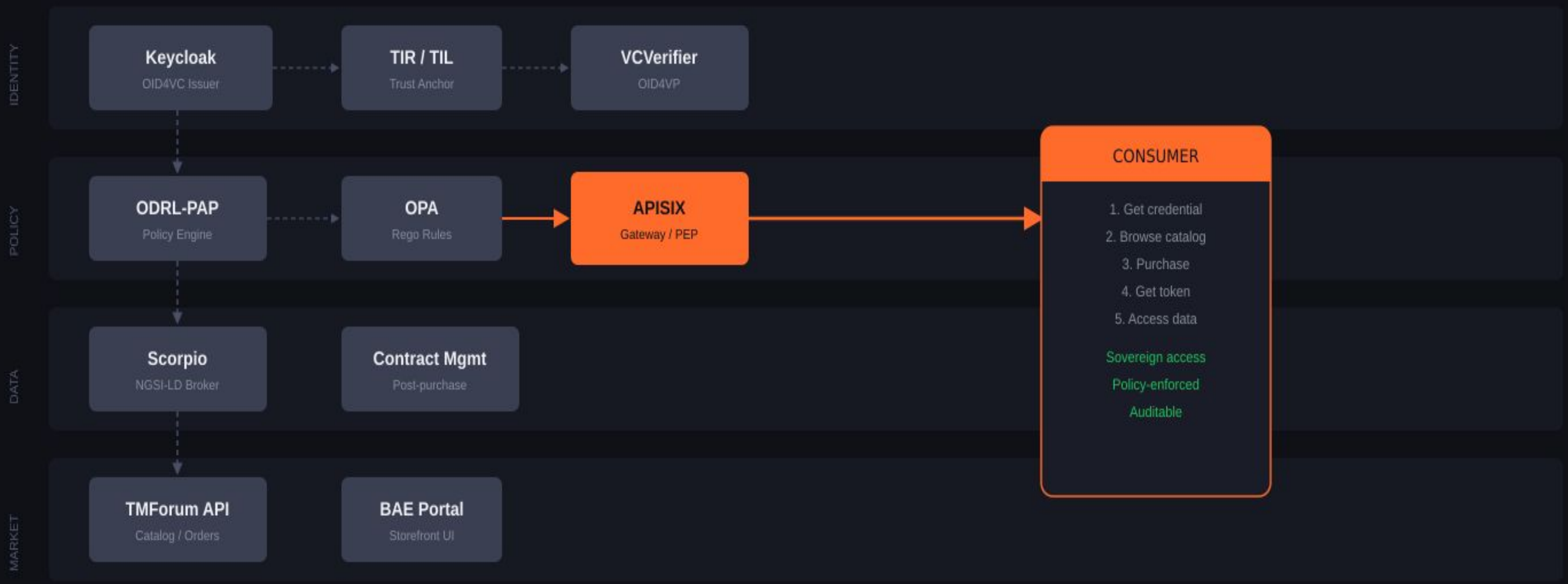
Multi-disciplinary teams. Reference architectures: IDS, Gaia-X, FIWARE. Looking beyond a single topic — processes and technology.

Deliverable

A working Marketplace inside a DataSpace, plus a tutorial on how to implement, design, and connect from scratch.

FIWARE Architecture

Four layers · Identity → Policy → Data → Market · Orange = the active enforcement path



From Signup to Data Access — 6 Steps

Grey = infrastructure. Orange = the live data access path (Steps 5–6 + Done). All steps validated on Fontys NetLab.

5 Deployment Phases

Each phase = one Maven profile from the official FIWARE doc/ directory.

#	Phase	Profile	Progress	Status
1	Local MVD	-Plocal	100%	COMPLETE
2	Central Marketplace	-Plocal,central	80%	1 BUG
3	Gaia-X Compliance	-Plocal,gaia-x	85%	EXT. DEP.
4	EDC Integration	-Plocal,dsp	0%	NOT STARTED
5	BAE + EUDI Wallet	Marketplace UI	0%	NOT STARTED

2 Open Blockers

Phase 2

Certificate Loading Bug

Fix →

Inspect consumer CM pod secret — likely wrong cert format

Phase 3 & 5

EUDI Wallet Required

Fix →

Install EUDI APK via Squid proxy — one setup unblocks both phases

Phases 4 + 5 upcoming: EDC cross-connector interoperability (-Plocal,dsp) · BAE graphical storefront with wallet login

Key Learnings

1

Networking precision matters

nip.io · br_netfilter · Squid — one wrong hostname breaks everything

2

Identity is the foundation

OID4VC/OID4VP flows through every single access decision

3

Policies must match purchases

Contract Management → TIR → PAP is the critical post-purchase chain

4

Keycloak 26 breaks curl flows

OID4VCI requires a real wallet — document this early

5

infra Traefik ≠ ingressClassName

did:web silently drops without a ConfigMap patch — very hard to debug

6

KUBECONFIG doesn't persist

Source it in every shell — add to .bashrc immediately

Deliverables & Conclusion

Deliverable

A working Marketplace inside a DataSpace + a tutorial on how to implement, design, and connect from scratch.

1	2	2	1
Phase complete	Phases mostly done	Phases documented	Blocker unblocks 2

Research Question — Answered

How can we implement and configure the Marketplace in a DataSpace?

Demonstrated a fully working Marketplace (TMForum + FIWARE) inside a DataSpace. Verified purchase-to-access flows. Documented 5 deployment phases from zero to Gaia-X compliance.

Built on: FIWARE DSC · TMForum Open APIs · OID4VC / OID4VP · ODRL + OPA · Gaia-X Trust Framework · Eclipse DSP (EDC)

Thank You

Questions & Discussion

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Repo: FIWARE Data Space Connector
Profiles: -Plocal · -Plocal,central · -Plocal,gaia-x · -Plocal,dsp